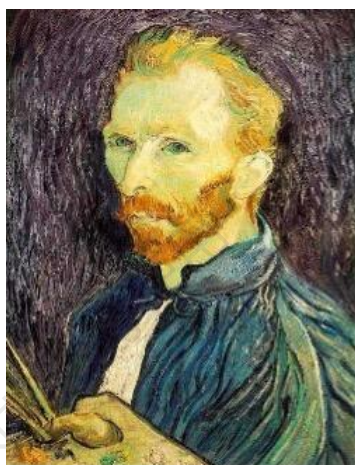


# Image Pyramids

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## Naïve image downsampling



1/2



1/4 (2x zoom)



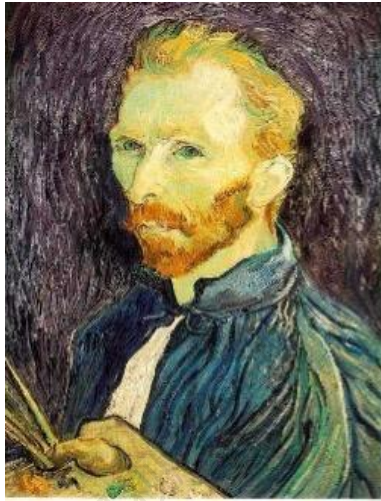
1/8 (4x zoom)

**Problems of Naïve image downsampling**

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## Better image downsampling



1/2

Apply a smoothing filter first, then throw away half the rows and columns

Gaussian filter  
delete even rows  
delete even columns



1/4

Gaussian filter  
delete even rows  
delete even columns

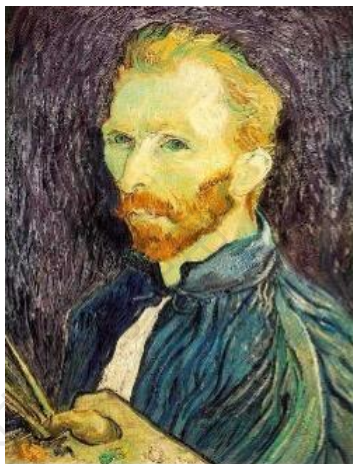


1/8

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## Better image downsampling



1/2



1/4 (2x zoom)



1/8 (4x zoom)

**A sequence of subsampled images**

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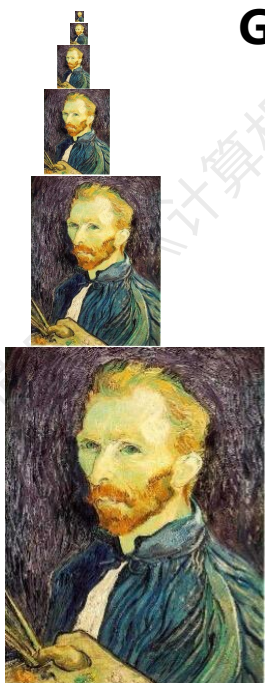
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# Gaussian Image Pyramid

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## Gaussian image pyramid



The name of this sequence of subsampled images

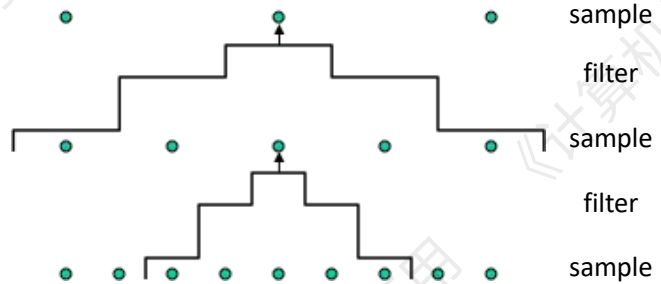
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# Constructing a Gaussian pyramid

## Algorithm

```
repeat:
  filter
  subsample
until min resolution reached
```



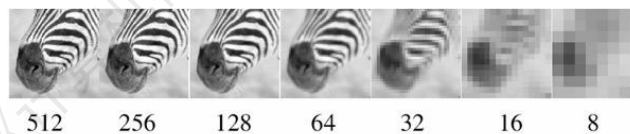
Question: How much bigger than the original image is the whole pyramid?

Answer: Just  $4/3$  times the size of the original image! (How did I come up with this number?)

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## Gaussian pyramid



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# Revisiting Sampling

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## The Nyquist-Shannon sampling theorem

A continuous signal can be perfectly reconstructed from its discrete version using linear interpolation, if sampling occurred with frequency:

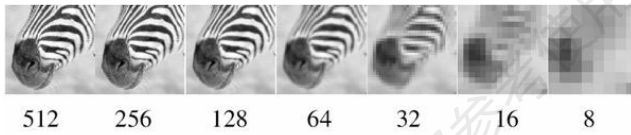
$$f_s \geq 2f_{\max} \quad \leftarrow \text{This is called the Nyquist frequency}$$

Equivalent reformulation: When downsampling, aliasing does not occur if samples are taken at the Nyquist frequency or higher.

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# Gaussian pyramid



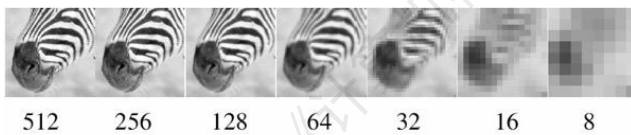
How does the Nyquist-Shannon theorem relate to the Gaussian pyramid?



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# Gaussian pyramid



How does the Nyquist-Shannon theorem relate to the Gaussian pyramid?

- Gaussian blurring is low-pass filtering.
- By blurring we try to sufficiently decrease the Nyquist frequency to avoid aliasing.



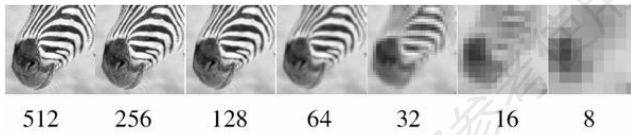
How large should the Gauss blur we use be?

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## Some properties of the Gaussian pyramid



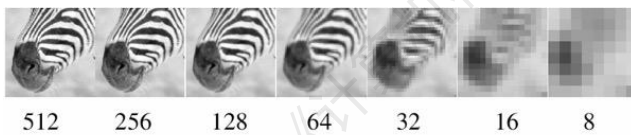
What happens to the details of the image?



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## Some properties of the Gaussian pyramid



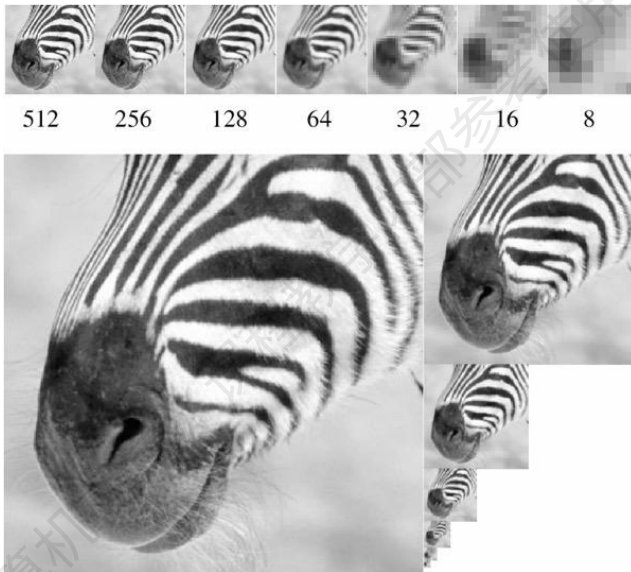
- What happens to the details of the image?
- They get **smoothed out as we move to higher levels.**
- What is preserved at the higher levels?



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## Some properties of the Gaussian pyramid

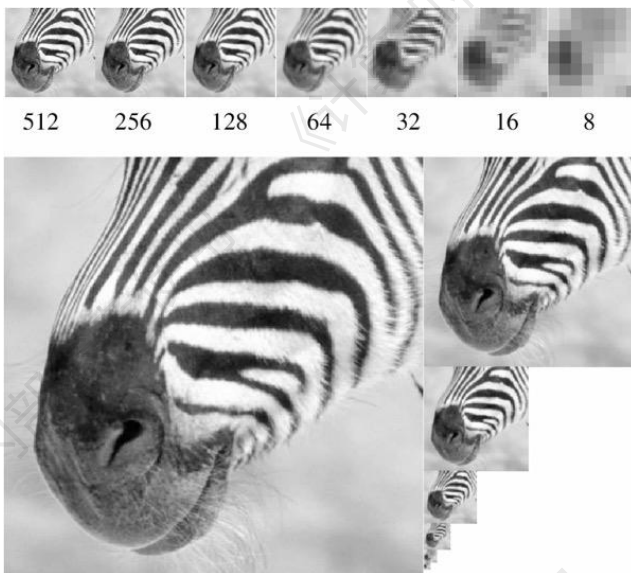


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- What happens to the details of the image?
  - They get smoothed out as we move to higher levels.
- What is preserved at the higher levels?
  - Mostly large uniform regions in the original image.
- **How would you reconstruct the original image from the image at the upper level?**

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## Some properties of the Gaussian pyramid



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- What happens to the details of the image?
  - They get smoothed out as we move to higher levels.
- What is preserved at the higher levels?
  - Mostly large uniform regions in the original image.
- How would you reconstruct the original image from the image at the upper level?
  - That's not possible.

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## Blurring is lossy



level 0



level 1 (before downsampling)

What does the residual look like?

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## Blurring is lossy



level 0



level 1 (before downsampling)



residual

Can we make a pyramid that is lossless?

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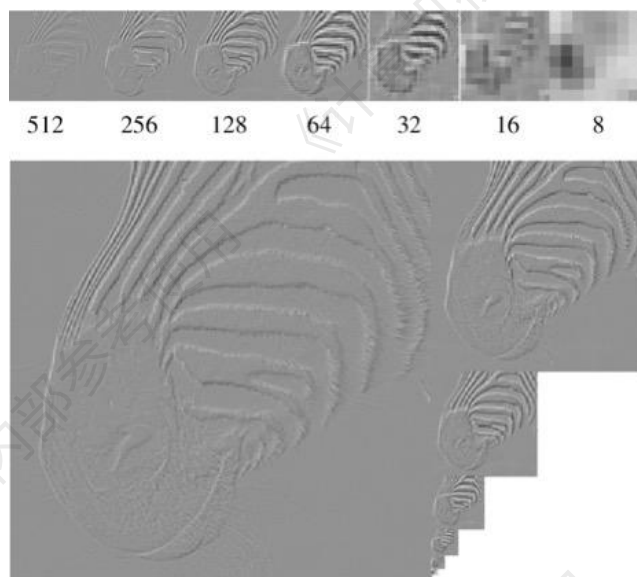
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# Laplacian image pyramid

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## Laplacian image pyramid



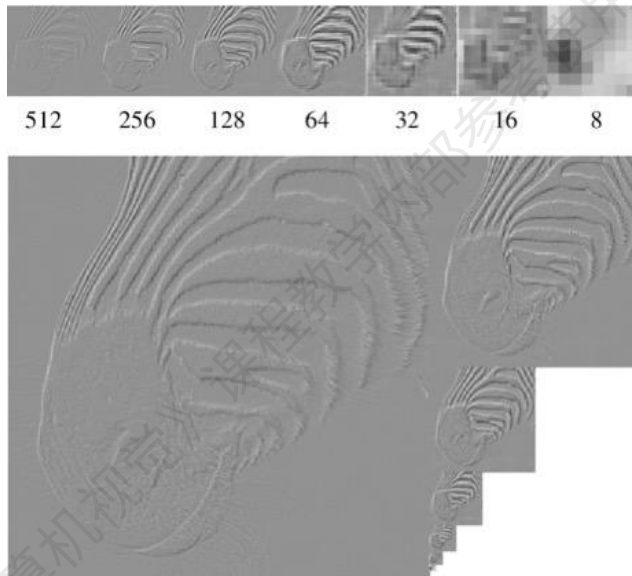
At each level, **retain the residuals instead of the blurred images themselves.**

Can we reconstruct the original image using the pyramid?

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## Laplacian image pyramid



At each level, retain the residuals instead of the blurred images themselves.

Can we reconstruct the original image using the pyramid?

- Yes we can!

**What do we need to store to be able to reconstruct the original image?**

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## Let's start by looking at just one level



level 0

=



level 1 (upsampled)

+



residual

Does this mean we need to store both residuals and the blurred copies of the original?

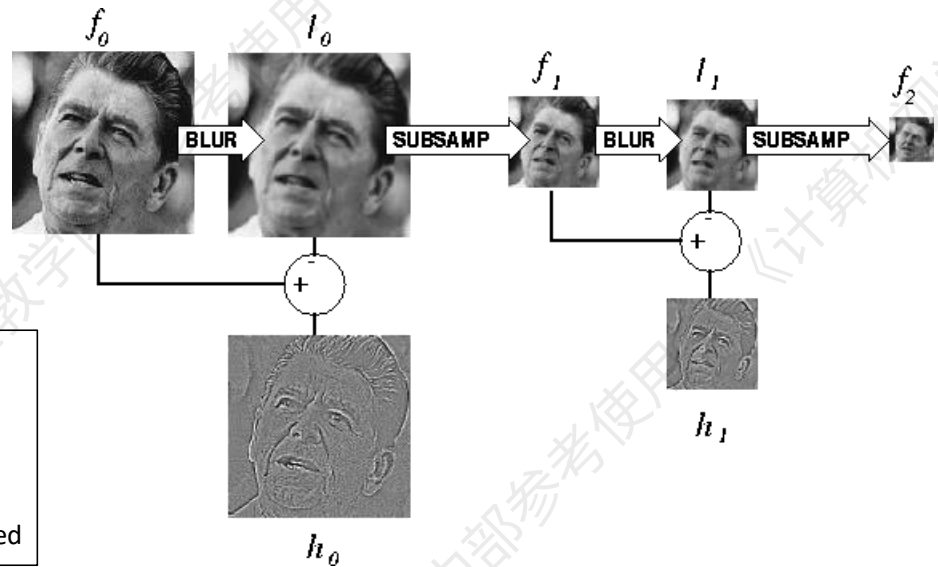
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## Constructing a Laplacian pyramid

### Algorithm

repeat:  
 filter  
 compute residual  
 subsample  
 until min resolution reached



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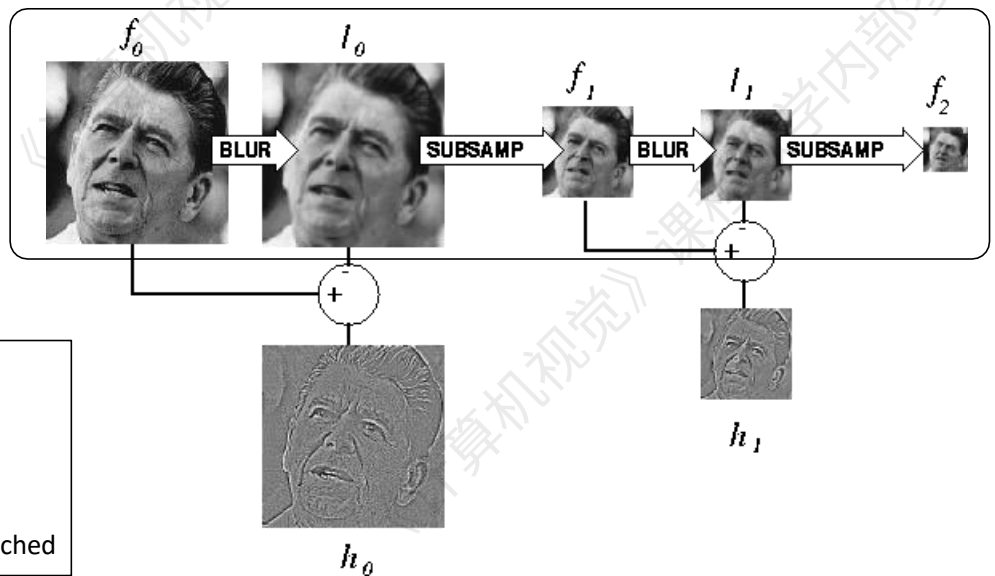
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## Constructing a Laplacian pyramid

What is this part?

### Algorithm

repeat:  
 filter  
 compute residual  
 subsample  
 until min resolution reached

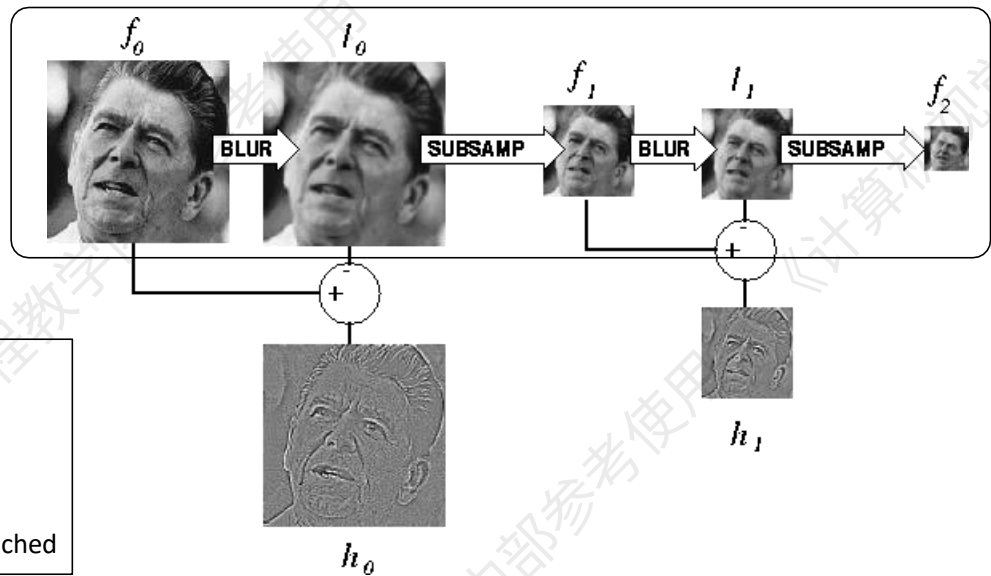


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## Constructing a Laplacian pyramid

It's a Gaussian pyramid.



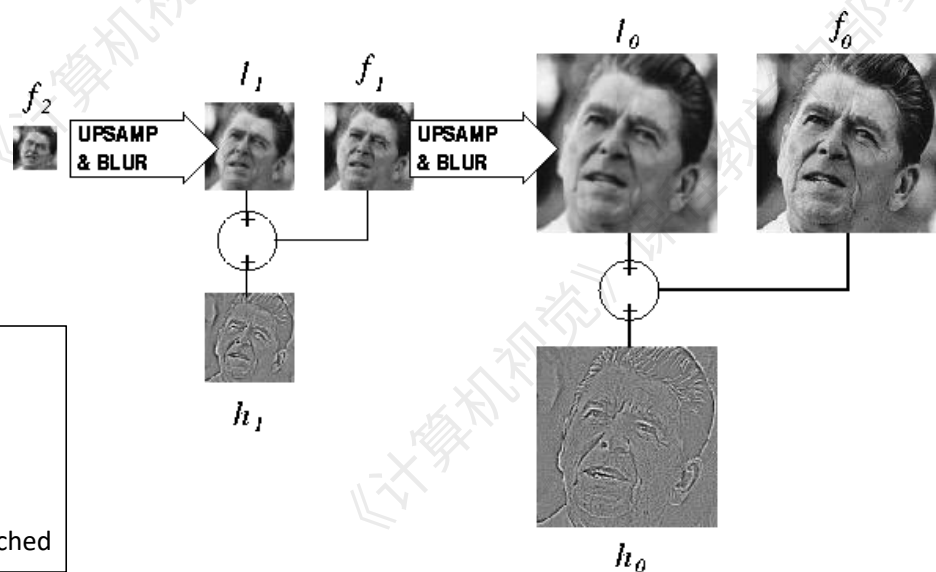
### Algorithm

repeat:  
 filter  
 compute residual  
 subsample  
 until min resolution reached

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## Reconstructing the original image



### Algorithm

repeat:  
 upsample  
 sum with residual  
 until orig resolution reached

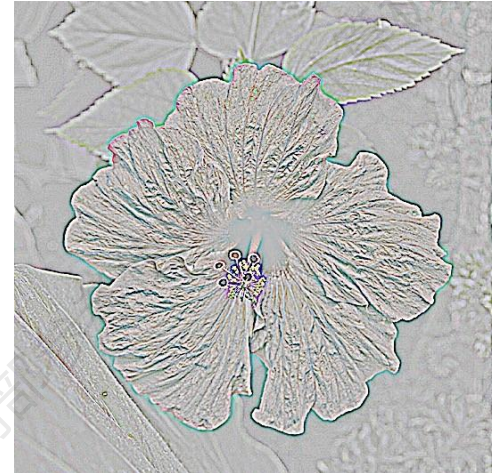
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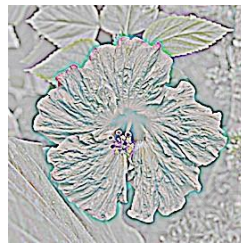
## Gaussian vs Laplacian Pyramid



Shown in opposite order for space.



Which one takes more space to store?



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## Why is it called a Laplacian pyramid?

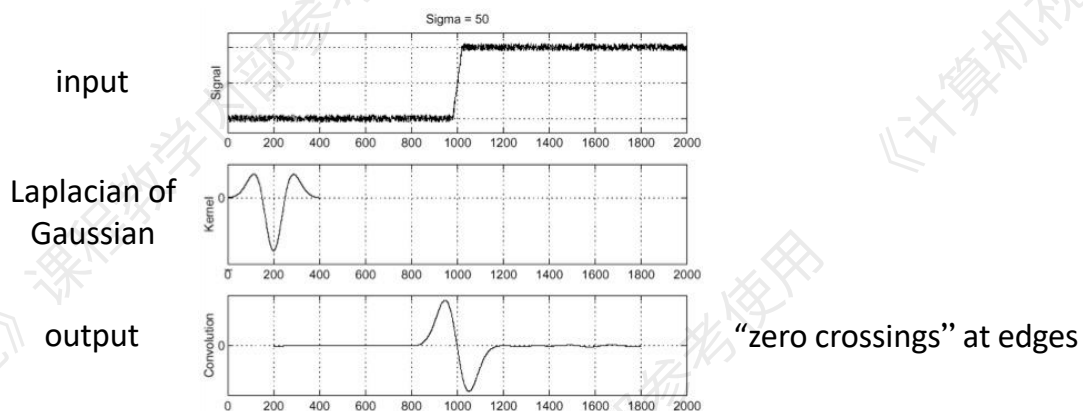
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## Reminder: Laplacian of Gaussian (LoG) filter

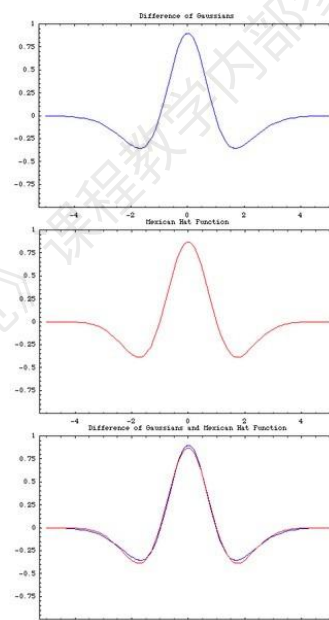
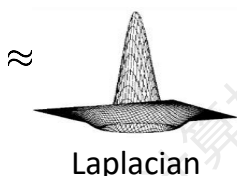
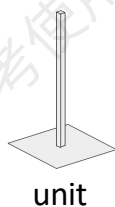
As with derivative, we can combine Laplace filtering with Gaussian filtering



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## Why is it called a Laplacian pyramid?



Difference of Gaussians approximates the Laplacian

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## Still used extensively



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## Still used extensively



foreground details enhanced, background details reduced



input image



user-provided mask

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## Other types of pyramids

Steerable pyramid: At each level keep multiple versions, one for each direction.



Wavelets: Huge area in image processing (see 18-793).



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## What are image pyramids used for?

image compression



multi-scale texture mapping

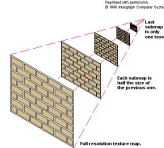
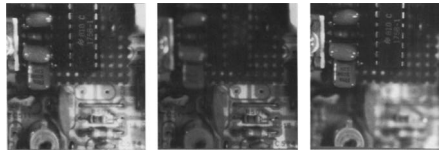


image blending



focal stack compositing



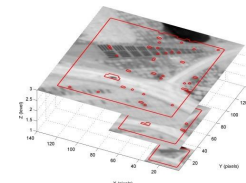
denoising



multi-scale detection



multi-scale registration

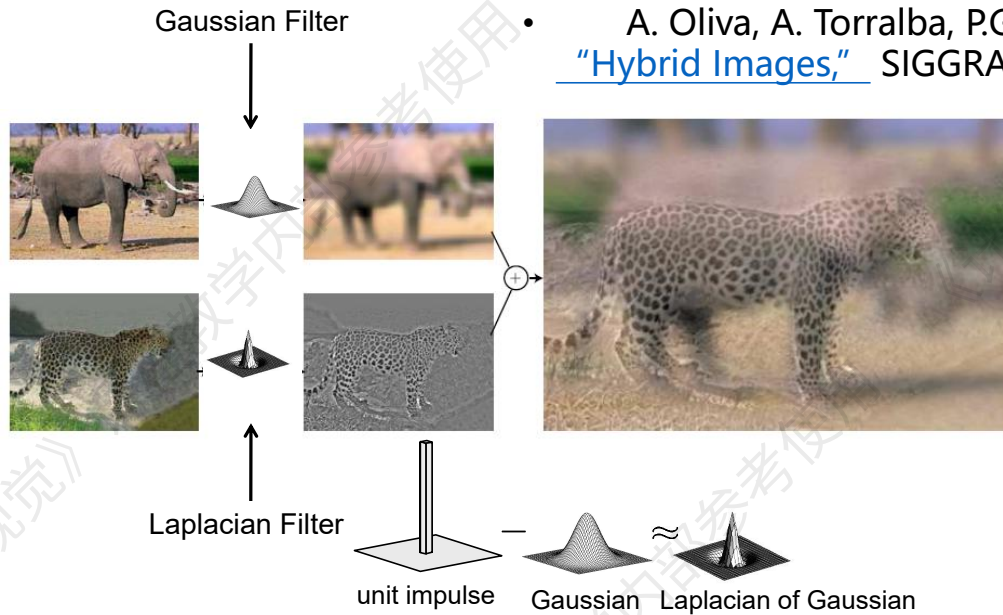


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## • Application: Hybrid Images



A. Oliva, A. Torralba, P.G. Schyns, ["Hybrid Images,"](#) SIGGRAPH 2006

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Salvador Dali invented Hybrid Images?



**Salvador Dali**  
*"Gala Contemplating the Mediterranean Sea, which at 20 meters becomes the portrait of Abraham Lincoln", 1976*

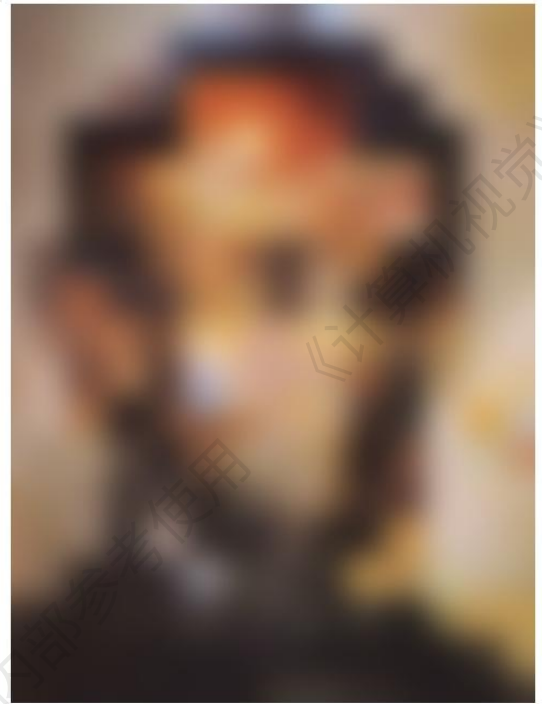
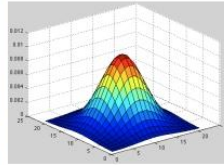
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Dali

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### Laplacian Pyramid Blending

Given a mask with black and white pixels only.

Given two input images, background image and foreground image.

Mask Image



Background Image

+



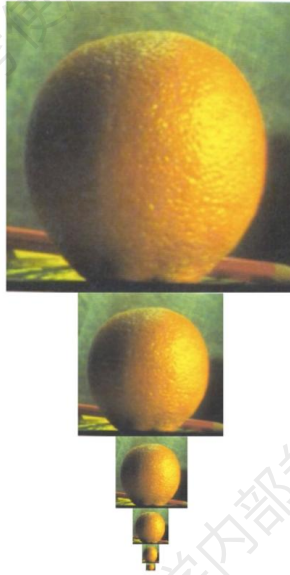
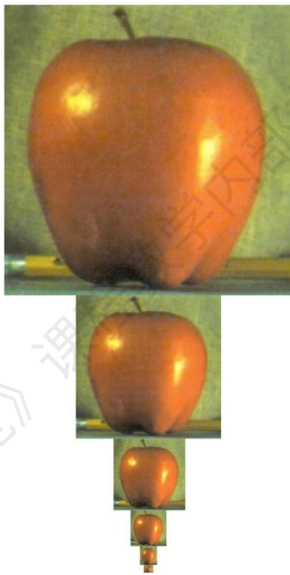
Foreground Image

=



Result

# Image Pyramids

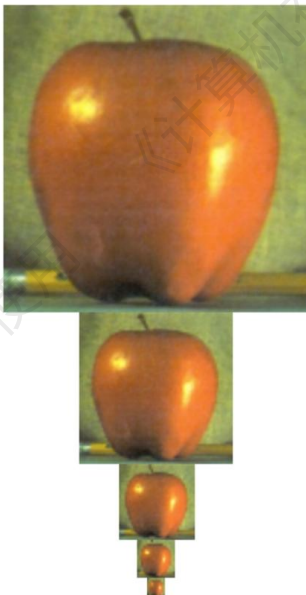


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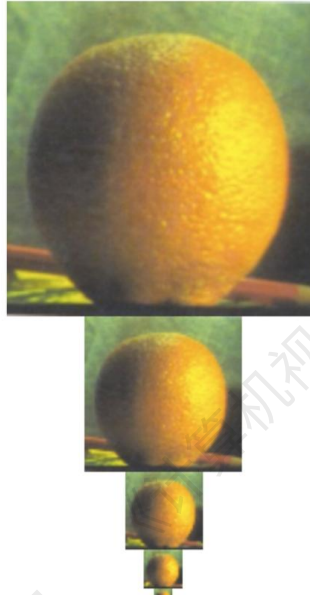
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## Gaussian Pyramids

Background Image



Foreground Image

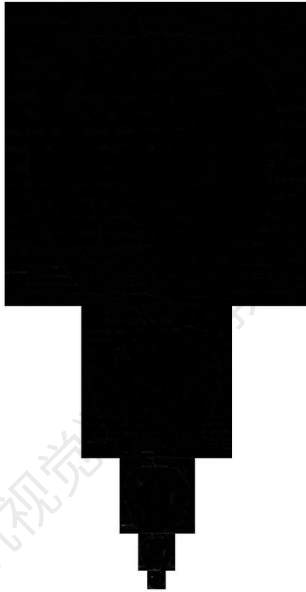


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# Laplacian Pyramids

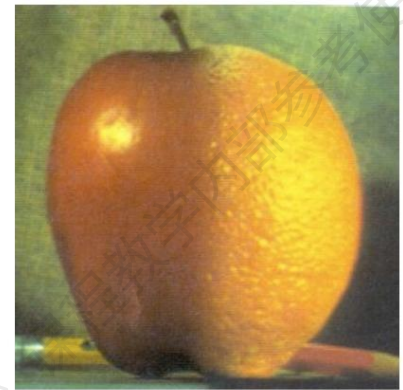
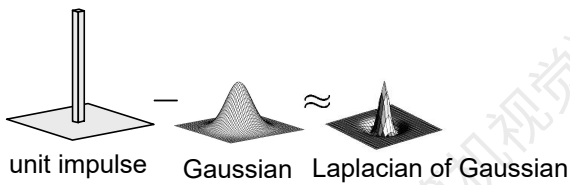
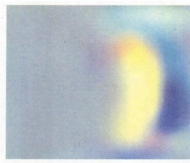
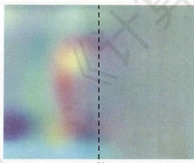
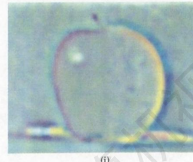
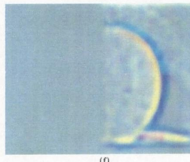
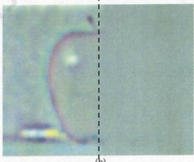
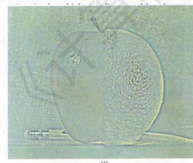
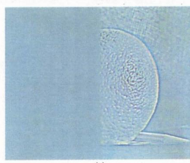
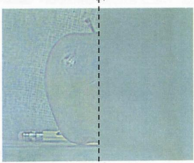
Background Image



Foreground Image



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©1999~2023 Integrative Intelligence and Learning Lab (I<sup>2</sup>L<sup>2</sup>)laplacian  
level  
4laplacian  
level  
2laplacian  
level  
0

left pyramid

right pyramid

blended pyramid

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# Frequency-domain filtering in human vision



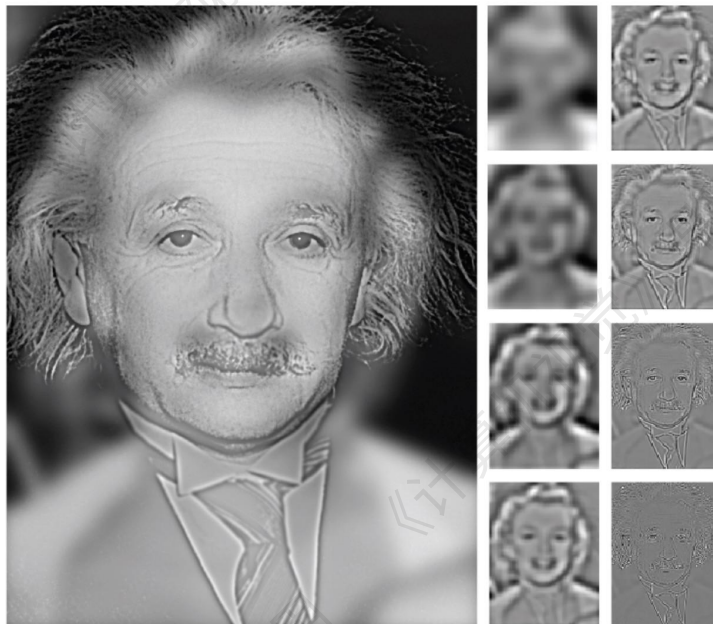
*"Hybrid image"*

Aude Oliva and Philippe Schyns

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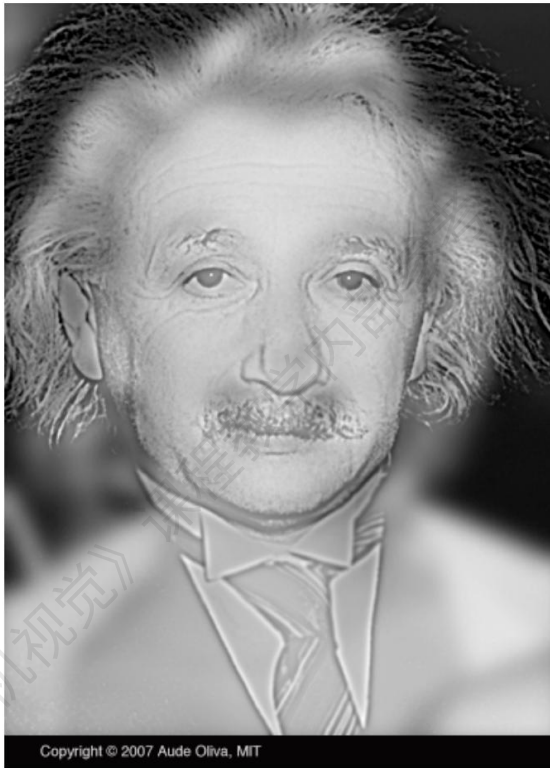
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## Frequency-domain filtering in human vision



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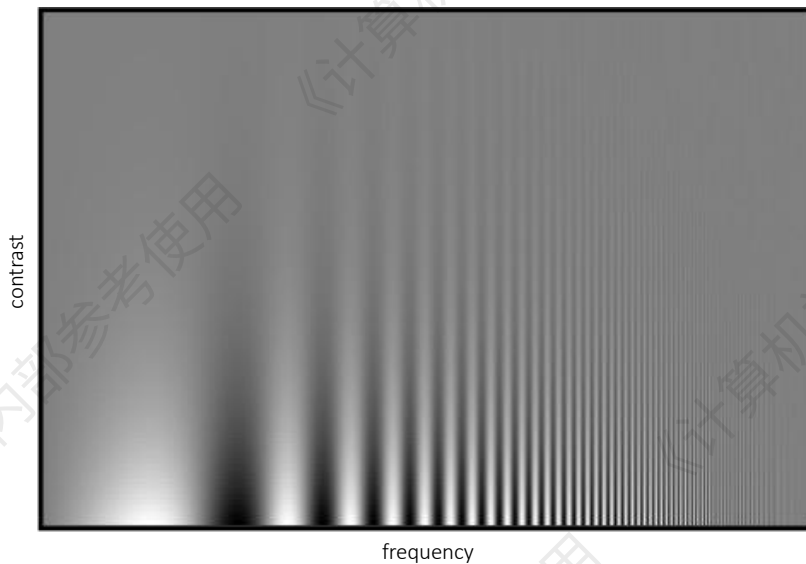
ML Center,

.47

ing Lab (I<sup>2</sup>L<sup>2</sup>)

## Variable frequency sensitivity

Experiment: Where do you see the stripes?

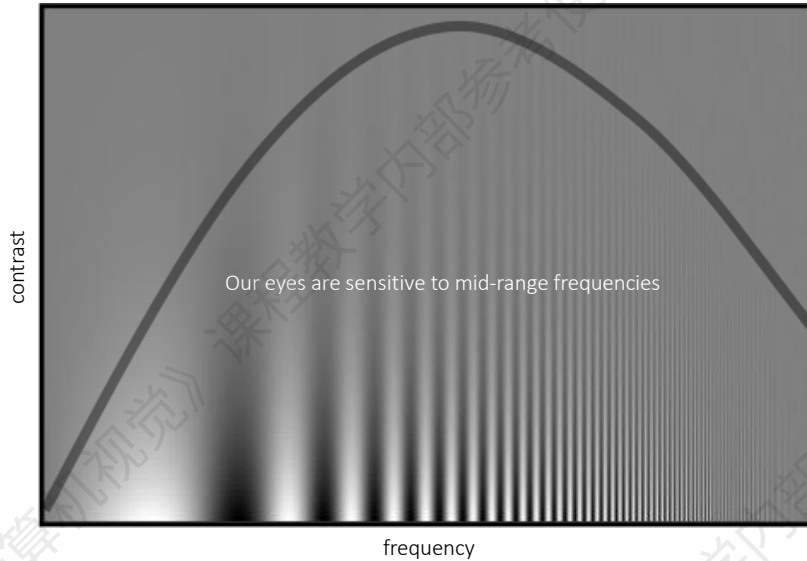


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# Variable frequency sensitivity

Campbell-Robson contrast sensitivity curve



- Early processing in humans filters for various orientations and scales of frequency
- Perceptual cues in the mid frequencies dominate perception

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